



JULIAN CABRERA MARCEGLIA

Manufacturing Systems & Digital Transformation Engineer | Industry 4.0 | Manufacturing Data Platforms | Automotive Manufacturing

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PROFESSIONAL SUMMARY

Manufacturing Systems professional with 21+ years of experience in automotive manufacturing environments, combining deep shop-floor knowledge, quality systems expertise and digital manufacturing capabilities.

Experienced in supporting Toyota manufacturing programs through the evolution from production operations into manufacturing information systems, industrial data management and digital transformation initiatives.

Specialized in understanding manufacturing process challenges, structuring operational information and developing digital solutions that improve traceability, visibility and engineering decision-making.

Designed and implemented manufacturing data solutions integrating multiple industrial information sources, transforming fragmented production information into structured datasets supporting quality improvement and operational analysis.

Developed an Industrial Measurement Data Platform that centralized more than 50,000 manufacturing records, improved reporting efficiency approximately 10x and created scalable foundations for manufacturing information management.

Combines automotive manufacturing experience with software engineering and industrial data capabilities to support manufacturing systems improvement, process optimization and digital transformation initiatives.

CORE EXPERTISE

- Manufacturing Systems & Digital Transformation
- Manufacturing Information Systems
- Digital Manufacturing Solutions
- Industry 4.0 Concepts

- Manufacturing Data Integration
- Production Information Management
- Manufacturing Traceability
- Process Digitalization
- Data-Driven Manufacturing Improvement
- Operational Visibility
- Manufacturing Operations
- Automotive Manufacturing
- Production Processes
- Quality Systems
- Process Improvement
- Continuous Improvement
- Root Cause Analysis
- Cross-functional Engineering Collaboration
- Manufacturing KPI Management
- Industrial Data & Analytics
- Industrial Data Platforms
- Engineering Data Management
- Manufacturing Analytics
- Reporting Automation
- Statistical Process Control
- Process Capability Analysis
- Data-Driven Decision Support

PROFESSIONAL EXPERIENCE

Metalsa

Quality Technician Supervisor

Sep 2016 – Present

Buenos Aires Province, Argentina

Support Toyota automotive manufacturing programs by combining manufacturing process knowledge, quality systems expertise and digital information capabilities.

Work at the intersection of production operations, engineering requirements and manufacturing data.

Manufacturing Systems & Digitalization Initiatives

- Designed and implemented a Manufacturing Measurement Data Platform integrating information from multiple industrial measurement sources.
- Structured manufacturing information flows to improve traceability, accessibility and engineering analysis.
- Integrated CMM, FARO and manual inspection information into a centralized engineering data environment.
- Managed more than 50,000 manufacturing measurement records.
- Automated reporting workflows, improving report generation efficiency approximately 10x.
- Developed scalable information management solutions supporting increasing product complexity.
- Improved access to manufacturing information for quality and engineering teams.

Manufacturing Quality & Process Improvement

- Lead KPI review meetings and cross-functional improvement activities involving production, quality and engineering teams.
 - Analyze manufacturing process performance using SPC, MSA, GR&R and capability studies.
 - Support root-cause analysis and corrective action initiatives.
 - Provide technical authority for dimensional acceptance and rejection decisions.
 - Develop and maintain measurement programs using PC-DMIS and FARO systems.
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Quality Technician

Sep 2008 – Sep 2016

Supported automotive manufacturing quality operations for Toyota programs.

- Generated manufacturing information through inspection and measurement systems.
 - Performed dimensional analysis and process capability studies.
 - Supported engineering teams in identifying manufacturing improvement opportunities.
 - Developed and maintained hundreds of inspection programs.
 - Worked with manufacturing data to improve quality decision-making.
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Production Welder

Sep 2004 – Sep 2008

- Performed automotive production activities within manufacturing environments.
- Developed practical understanding of production processes, assembly operations and shop-floor requirements.

SELECTED DIGITAL TRANSFORMATION PROJECTS

Manufacturing Measurement Data Platform

Manufacturing Information System | Industrial Data Integration

Designed and implemented a digital manufacturing solution to improve production information management.

Manufacturing Challenge

Manufacturing information was generated from different sources, creating challenges related to:

- Data accessibility.
- Traceability.
- Reporting efficiency.

- Engineering analysis.

Solution

Developed a centralized platform integrating:

- CMM measurement data.
- FARO inspection information.
- Manual inspection records.
- Structured manufacturing datasets.
- Automated reporting workflows.

Impact

- Centralized 50,000+ manufacturing records.
- Improved manufacturing information visibility.
- Reduced reporting effort approximately 10x.
- Enabled scalable engineering analysis.

Real-Time Quality Prioritization System

Manufacturing Analytics | Quality Decision Support

Developed a manufacturing analytics solution to transform quality information into prioritized engineering actions.

Capabilities

- Automated data processing.
- Manufacturing performance analysis.
- Quality risk identification.
- Engineering decision support.

Impact

- Improved identification of manufacturing issues.
- Reduced manual analysis effort.
- Supported proactive quality improvement.

TECHNICAL CAPABILITIES

Digital Manufacturing	Industrial Data & Analytics	Manufacturing Engineering
• Manufacturing Information Systems • Digital Manufacturing • Industry 4.0 • Manufacturing Data Platforms • Engineering Information Management • Quality Systems	• Python • SQL • DuckDB • Pandas • NumPy • SciPy • Database Design • Reporting Automation • Data Analysis	• Automotive Manufacturing • Dimensional Engineering • CMM Systems • FARO Measurement • Mitutoyo Cosmos • Systems • PC-DMIS • GD&T • Statistical Process Control

PROFESSIONAL POSITIONING

Manufacturing Systems Engineer combining automotive manufacturing experience, industrial data capabilities and software engineering skills.

Experienced in connecting manufacturing operations, engineering teams and digital solutions to improve production visibility, traceability and decision-making.

Focused on enabling more connected, efficient and data-driven manufacturing environments.